□ Salesforce Summer Program – Day 16 (4–5 Hours)

□ Goal for Today

Today students learn:

"How developers diagnose, debug, improve, and maintain enterprise systems."

Focus:

- Debugging
- Developer tools
- Error analysis
- Performance thinking
- LWC best practices
- Maintainable architecture

This fills important remaining modules from the Trailmix.

□ Modules To Complete

1□ Find and Fix Bugs with Apex Replay Debugger

Focus:

- Debug logs
- Replay debugging
- Error tracing
- Root cause analysis

2□ Developer Console Basics

Focus:

- Query editor
- Logs
- Debugging tools

- Apex execution

3️⃣ Best Practices in Lightning Web Components

Focus:

- Performance
- Reusability
- Clean architecture
- Maintainability

Videos (Properly Verified Working Links)

1️⃣ Salesforce Apex: How to Read & Use Debug Logs for Troubleshooting

https://www.youtube.com/watch?v=tPPv4VQY89k

Focus:

- Debug logs
- Troubleshooting
- Error tracing
- Root cause analysis

Channel:
Apex Hours

2️⃣ Developer Console in Salesforce

https://www.youtube.com/watch?v=2KRGa86nbOg

Focus:

- Developer Console basics
- Query editor
- Executing Apex
- Debugging tools

Channel:
Salesforce Hulk

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3 LWC Best Practices

https://www.youtube.com/watch?v=q3e0Qx2D6lY

Focus:

- Clean components
- Performance optimization
- Reusable architecture

Channel:
Salesforce Developers

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4 Salesforce Developer Tutorial – Apex Replay Debugger

https://www.youtube.com/watch?v=iWXvnylWuR8

Focus:

- Replay debugger
- VS Code debugging
- Apex troubleshooting
- Enterprise debugging workflow

Channel:
Coding With The Force

CORE TASKS

Task 1 – Bug Analysis

Suppose:

- duplicate notifications occur
- attendance calculations wrong
- flow not triggering
- approval process stuck

Explain:
□ how would you debug each issue?

Task 2 □ – Performance Thinking

Suppose:
50,000 users use your system simultaneously.

What problems might occur in:

- UI
- backend
- database
- notifications
- automation

Task 3 □ – Maintainability Thinking

Answer:

□ Why should developers write:

- modular code
- reusable components
- debuggable systems
  instead of quick hacks?

Task 4 □ – Reflection

Answer:

□ Why is debugging one of the most important skills in software engineering?

□ GitHub Submission

Create:

/day16-debugging-best-practices/

README.md should include:

- Common bug scenarios
- Debugging approach
- Performance discussion
- LWC best practices
- Reflection

✍□ Revision Questions

1. Why are debug logs important?
2. Why is debugging difficult in enterprise systems?
3. What problems happen when systems scale?
4. Why should components be reusable?
5. Why is maintainability important?
6. Why should developers avoid tightly coupled code?
7. Why do enterprise systems require monitoring?
8. Why is troubleshooting an important engineering skill?

□□ Rules

- Think like a software engineer
- Focus on reliability and maintainability
- Use real-world reasoning
- Do NOT memorize tools blindly
- Spend around 4–5 hours maximum

□ End of Day Outcome

Students should now clearly understand:

□ Enterprise debugging workflow
□ Developer Console usage
□ Importance of maintainable architecture
□ LWC best practices
□ Reliability and troubleshooting thinking
